

Password Strength Evaluation Report

Task 6 – Create a Strong Password and Evaluate Its Strength

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1. Objective

To understand the characteristics of strong passwords by evaluating passwords of varying complexity using an online password strength checker.

2. Tool Used

- PasswordMeter (passwordmeter.com)
- Web Browser

3. Procedure

1. Opened PasswordMeter.
2. Created multiple passwords with different complexity levels.
3. Tested each password using the password strength checker.
4. Observed the strength rating, estimated crack time, and security feedback.
5. Compared the results and identified characteristics of strong passwords.

4. Password Evaluation Summary

Password Type	Strength	Observation	Estimated Security
Common Password	Very Weak	Commonly used password; cracked instantly.	Poor
Simple Alphanumeric	Weak	Contains keyboard pattern and short length.	Low
Medium Complexity	Fair	Good length and all character types, but includes a predictable year.	Moderate
Strong Complex Password	Strong	Long password with uppercase, lowercase, numbers and symbols.	Excellent

5. Findings

- Longer passwords significantly increase resistance to brute-force attacks.
- Using uppercase, lowercase, numbers, and special characters improves password strength.
- Avoid common passwords, keyboard patterns, and predictable years.
- Password managers and Multi-Factor Authentication (MFA) further improve account security.

6. Common Password Attacks

- Brute Force Attack – Tries every possible combination.
- Dictionary Attack – Uses common passwords and dictionary words.
- Credential Stuffing – Reuses leaked credentials on multiple websites.

7. Conclusion

The evaluation demonstrated that password strength depends on length, complexity, and unpredictability. Strong passwords containing a mix of character types and sufficient length require significantly more time to crack and provide better protection against common password attacks.